

Semester-V

Course Title	Operating Systems	L	T	P	C	Periods
Course Code	UCSH - 501	4	0	0	4	56
Course Category	DSC					
Course Objective: To give a basic idea of an OS and its functionalities. To give an in- depth understanding of CPU and its scheduling, memory management, and device management. This course covers other functionalities of OS such as file systems, virtual memory, disk request scheduling, concurrent processes, deadlocks, security, and integrity.						
Course Outcome: Develop the skill set to 1. To become familiar with the working of the general operating system. 2. To give perspectives to understand the OS services. 3. To know the working of the operating system processes management. 4. To identify and understand memory subsystems. 5. To know the basic file management system. 6. To understand the storage structure.						
Prerequisites for the Course: NA						
Unit	Topics	Contents				Periods
1	Introduction	What is an Operating System, Simple Batch Systems, Multiprogrammed Batches systems, Time-Sharing Systems, Personal -computer systems, Parallel systems, Distributed Systems, Real -Time Systems, Computing Environments, Computer System operation, I/O Structure, Storage structure and Hierarchy, Operating System components and Services- System calls, System Programs, System Structure				12
2	Processes	Process Concept, Process Scheduling, Operation on Processes CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Multiple-Processor Scheduling, Process Synchronization: Background, The Critical -Section Problem, Synchronization Hardware, Semaphores, Classical Problems of Synchronization				12

3	Deadlocks	System Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock.	10
4	Memory Management	Background, Logical versus Physical Address space, swapping, Contiguous allocation, Paging, Segmentation, Virtual Memory: Demand Paging, Page Replacement, Page-replacement Algorithms, Performance of Demand Paging, Allocation of Frames, Thrashing, Other Considerations.	12
5	Storage Management	File system Implementation, Directory Implementation, Allocation methods, free space management, Disk Structure, Disk Scheduling, Disk Management, Swap-Space Management, Disk Reliability	10
		Total	56
Reference: Silberschatz and Galvin, “Operating System Concepts”, Wiley-India, International Student Version, 8th Ed., 2010 Chapters: Text 1: 1, 3 (3.1, 3.2, 3.3), 5 (5.1, 5.2,5.3, 5.5), 6 (6.1, 6.2, 6.4, 6.5, 6.6), 7, 8 (8.1, 8.2, 8.3, 8.4, 8.6), 9 (9.2,9.4,9.5,9.6,9.9), 11(11.2-11.5), 12(12.2,12.4-12.6).			

Course Title	Computer Networks	L	T	P	C	Periods
Course Code	UCSH-502	4	0	0	4	56
Course Category	DSC					
Course Objective	1. To develop an understanding of modern network architectures from design and performance perspective. To introduce the student to the major concepts involved in wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs). 2. To provide an opportunity to do network programming To provide WLAN measurement ideas.					
Course Outcomes: Develop the skill set to 1. Explain the functions of the different layers of the OSI Protocol. 2. Draw the functional block diagram of wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs) to describe the function of each block. 3. For a given requirement (small scale) of wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs) design it based on the market available component 4. For a given problem related TCP/IP protocol developed the network programming. 5. Configure DNS DDNS, TELNET, EMAIL, File Transfer Protocol (FTP), WWW, HTTP, SNMP, Bluetooth, Firewalls using open source available software and tools.						
Prerequisites for the Course: UCSH-401						
Unit	Topic	Contents				Periods
1	Introduction	Introduction to computer networks, evolution of computer networks and its uses, Advantages and Disadvantages of Computer Network, reference models: OSI reference Models, TCP/IP Protocol Suite Networking fundamentals: Internet, Circuit switching vs Packet switching, ISPs, Delay and Loss in Packet Switched Networks				10
2	Local Area Network	LAN Architecture, LAN topologies- Bus/ Tree LAN, Ring LAN, Star LAN, Wireless LAN, Ethernet and Fast Ethernet, Token Ring				8
3	Application layer and data link layer	Application Layer Protocols: HTTP, FTP, SMTP, DNS Data link layer design issues, Flow Control- Stop and Wait, Error Detection, Error Control, error				10

		detection and correction, data link layer protocols, sliding window protocols, example of data link protocol- HDLC	
4	Medium access layer	Channel allocation problem, multiple access protocols, Introduction to ALOHA, CSMA/CD, CSMA/CA	8
5	The network layer	Introduction, Routers, Network layer concepts, shortest path routing, flooding, distance vector routing, link state routing (without algorithms), congestion control and quality of service, internetworking, IP, Ipv4 Addressing vs Ipv6	10
6	The transport layer	The transport layer services, elements of transport protocols, TCP and UDP, Brief introduction to presentation and session layer, E-mail	10
		Total	56
Reference: 1. Data Communication & networking: Forouzan, B. A. , Vth Ed, The McGraw-Hill Companies, Inc - 2017. 2. Data And Computer Communications, by William Stallings, VII edition, Pearson Education, 2005.			
Suggested Readings: 1. Andrew S. Tanenbaum, Computer Networks, Fourth Edition, Pearson Education, 2003 2. Jim Kurose, Keith Ross, A Top Down Approach Featuring the Internet , Third Edition, Pearson Education, 2004			

Course Title	Software Lab in Java	L	T	P	C	Periods
Course Code	UCSH-503	0	0	6	3	84
Course Category	LW					
Course Objectives: This course introduces the basics of Java programming.						
Course Outcome: Develop the skill set to 1. Write java code for basic programming. 2. Understand and implement object oriented concepts like Abstraction, Encapsulation and Inheritance 3. Implement various data structures like List, Tree and Graph 4. Understand related concepts in Java like Exceptions, Packages 5. Implement simple programs with graphics in Java						
Prerequisites for the Course: NA						
Unit	Topic	Contents				Periods
1	Primitives and Objects in Java	Simple programs for learning primitive data, basics of creating, using objects in java				10
2	Loops and Control structures	Simple programs using if-else, for, while, switch-case, break, continue				10
3	Access modifiers and Constructors	Private, default, protected and public access modifiers for fields and methods. Default Constructor, Constructor invocations, static, abstract, final				15
4	Inheritance	Object class, super, Multi-level inheritance, Typecasting, Constructors in Inheritance, Fields and Methods in Inheritance, Overriding, Final & Abstract classes				15
5	Arrays, Exceptions, Packages, Interfaces	1D and 2D arrays in Java, Inheritance in Arrays, Checked and Unchecked Exceptions, try, catch, throw, finally, Different ways to import packages, Class path, Inheritance in Packages, Using interfaces for multiple inheritance				14

6	Graphic, sets, and Process	Swing, Collection, Multithreading	20
		Total	84
Reference & Suggested Readings: 1. Herbert Schildt, Java: The Complete Reference, , McGraw-Hill Education, 12th edition, 2021, ISBN-13: 978-1260463415. 2. Cay S. Horstmann, Core Java Volume I – Fundamentals, 12th Edition, Pearson Education, 2021 ISBN-13: 978-0137673620 3. Kathy Sierra & Bert Bates, Head First Java, O'Reilly Media, 2005, ISBN-13: 978-059600920			

Course Title	Optimization for Machine Learning	L	T	P	C	Periods
Course Code	UDSC-501	3	0	2	4	70
Course Category	IDM					
Course Objective: To modeling and discussion of documented real-world applications. Study of mathematical programming algorithms for optimization. Apply the mathematical results and numerical techniques of optimization theory to concrete Engineering problems.						
Course Outcome: Develop the skill set to 1. discovery, study and solve optimization problems; 2. Investigate, study, develop, organize and promote innovative solutions for various applications. 3. Apply mathematical skills to model optimization problems. 4. Critically analyze theoretical principles and choose the relevant optimization techniques for a specific problem. 5. Exposed to solutions for various types of optimization problems and will be enabled to adopt them for the situation at hand.						
Prerequisites for the Course: 10+2 Mathematics, Python programming, UDSC-101 Calculus						
Unit	Topic	Description				Periods
1	Introduction	Mathematical Review – Lines – Hyperplanes and linear varieties – convex sets – neighborhoods – polytopes and polyhedra – derivative matrix – level sets and gradients-Taylor’s series				6
2	Unconstrained Optimization	Unconstrained Optimization – conditions for local minimizers – one-dimensional search methods including golden section, Fibonacci, bisections and Newton’s method – Gradient methods including steepest descent methods, Newton's method and its analysis.				8
3	Linear Programing	Modeling with Linear Programming The Simplex method and Duality				8
4	Non-Linear Constrained Optimization	Non-linear constrained optimization – problems with equality constraints – Lagrange condition – second-order condition – minimizing quadratics subject to equality constraints – problems with inequality constraints – Karush-Kuhn-Tucker conditions – second-order conditions.				8

5	Convex Optimization	Convex Optimization Problems - Convex Functions - Convex Optimization Problems	6
6	Machine Learning view for Optimization	The minutiae of Gradient Descent, Properties of Optimization in ML, Computing Gradient wrt vectors, Linear Regression and Optimization models for Binary Targets	6
		Total	42

Suggested/Additional Readings:

1. L.R. Foulds, Optimization Techniques, Springer, Utm, 1981
2. <http://freevideolectures.com/Course/3072/Numerical-Optimization>
3. Hamdy A. Taha, Operations Research, Pearson Publication, 9th edition, (2014).
4. Boyd, Stephen, and Lieven Vanderberghe, Convex Optimization. Cambridge, UK: Cambridge University Press, 2004.

UDSC-501: Practicals - Optimization for Machine Learning

Unit	Topics	Contents	Periods
1	Unconstrained Optimization	Implementation of Unconstrained Optimization using Python: Newton-Raphson method; Implementation of Gradient Ascent and Gradient Descent methods	10
2	Solving LPP	Solving LP problems using Excel solver; Sensitivity Analysis using Excel solver; Solving LP problems using <i>scipy.optimize.linprog</i>	10
3	Python Library <i>scipy.optimize</i>	Univariate function minimization; Constrained and Unconstrained minimization of multivariate scalar function; Global Optimization; Least-Square Optimization; Custom minimizers	8
		Total	28

References:

1. SciPy Reference Guide, Retrieved July 7, 2022 from <https://docs.scipy.org/doc/scipy-1.7.1/scipy-ref-1.7.1.pdf>
Section 2.4 (Optimization - *scipy.optimize*)
2. Christian Hill, Learning Scientific Programming with Python, Second Edition, Cambridge

University Press, 2020

3. Edwin K. P. Chong and Stanislaw H. Zak, An Introduction to Optimization, Fourth Edition, Wiley Inter - Science Publication, John Wiley and Sons Inc., 2013.

4. Hamdy A.Taha, Operations Research- An Introduction, 10th Edition, Pearson Education - 2017

Course Title	Data Mining and Machine Learning	L	T	P	C	Periods
Course Code	UDSC-502	3	0	2	4	70
Course Category	IDM					
Course Objectives: 1. To introduce students to the basic concepts and techniques of Machine Learning. 2. To become familiar with regression methods, classification methods, Pattern Mining and clustering methods. 3. To become familiar with Kernel Methods and Dimensionality reduction Techniques						
Course Outcomes: Develop the skill set to • Gain knowledge about basic concepts of Machine Learning • Identify machine learning techniques suitable for a given problem • Solve the problems using various machine learning techniques • Apply Dimensionality reduction techniques						
Prerequisites: UDSC-101/UMAT-101, UDSC-102, UDSC-201, UDSC-203 or Equivalent						
Unit	Title	Contents			Periods	
1	Introduction	Data Matrix, Numeric Attributes, Categorical Attributes, Graph Data, Kernel Methods, High-dimensional Data, Dimensionality Reduction, Types of Machine Learning			8	
2	Regression	Linear Regression, Logistic Regression, Neural Networks, Regression Evaluation			8	
3	Classification	Probabilistic Classification, Decision Tree Classifier, Linear Discriminant Analysis, Support Vector Machines, Classification			12	

		Assessment.	
4	Clustering	Representative-based Clustering, Hierarchical Clustering, Density-based Clustering, Spectral and Graph Clustering, Clustering Validation	14
5	Frequent pattern mining	Itemset Mining, Summarizing Itemsets Sequence Mining, Graph Pattern Mining.	14
Total			56

Key Text(s):

1. Mohammed J. Zaki, Wagner Meira, Jr., Data Mining and Machine Learning: Fundamental Concepts and Algorithms, 2nd Edition, Cambridge University Press, March 2020.
2. Trevor Hastie, Robert Tibshirani, and Jerome H. Friedman, The Elements of Statistical Learning, Second Edition, Springer, 2009
Retrieved on July 05, 2022 from <https://hastie.su.domains/Papers/ESLII.pdf>
4. Christopher M Bishop, Pattern Recognition and Machine Learning, Springer, 2006
5. Rogers and Girolami, A First Course in Machine Learning, Chapman and Hall/CRC, 2015

References:

1. Barber, Bayesian Reasoning and Machine Learning, Cambridge University Press, 2012
2. Hal Daumé III, A Course in Machine Learning, e-Edition, 2012
3. Mitchell, Machine Learning, McGraw Hill, 1997
4. CS229-Machine Learning, Stanford University, Retrieved June 30, 2022 from <https://see.stanford.edu/Course/CS229>

UDSC-502: Practicals- Data Mining and Machine Learning

Unit	Title	Contents	Periods
1	Introduction	Review numpy, pandas and matplotlib for statistical, Probabilistic, algebraic and geometric views of data matrix Implementation of Linear algorithms, Non-Linear Algorithms and ensembling algorithms, bias-variance tradeoff, from scratch in python	12
2	Python Library scikit-learn	Develop different ML algorithms using scikit-learn	16

Key Text(s):

1. Jason Brownlee, Master Machine Learning Algorithms, Discover How They Work and Implement them from Scratch, Machine Learning Mastery, eBook, 2017
Retrieved June 30, 2022
<https://github.com/AmandaZou/master-machine-learning-algorithms>
2. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'reilly, 2019
3. Scikit-learn Documentation, retrieved June 30, 2022 from [scikit-learn Tutorials — scikit-learn 1.0.2 documentation](#)